

Ip addressing and subnetting

1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.



Class	IP Address Range	Used For	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	Large networks	Google or Microsoft company network
Class B	128.0.0.0 – 191.255.255.255	Medium-sized networks	University or college network
Class C	192.0.0.0 – 223.255.255.255	Small networks	Home Wi-Fi router (192.168.1.1)
Class D	224.0.0.0 – 239.255.255.255	Multicast	Live YouTube streaming or IPTV
Class E	240.0.0.0 – 255.255.255.255	Experimental/Research	Network research and testing

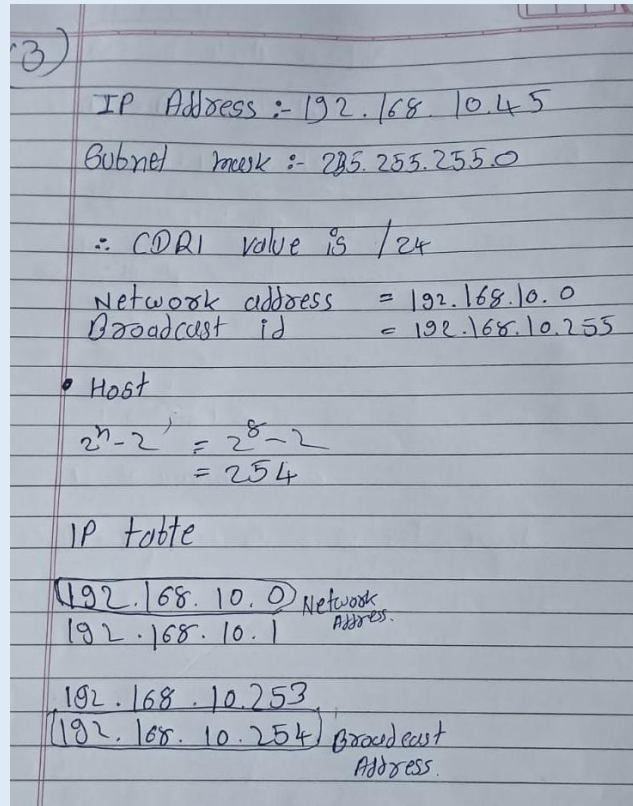
2. A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.



- Your IP address 192.168.1.300 is invalid because each octet in an IPv4 address can only have a value from 0 to 255. The number 300 is outside the valid range, so the device cannot connect to the network.
- For the 192.168.1.0/24 subnet, a valid host IP address would be between 192.168.1.1 and 192.168.1.254 (assuming .0 is the network address and .255 is the broadcast address).

3. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint: Show your working steps for each calculation.



4. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network



Group	Network Address	Subnet Mask	CIDR	Host Range
Delivery Partners	10.0.0.0	255.255.255.192	/26	10.0.0.1 – 10.0.0.62
Restaurants	10.0.0.64	255.255.255.192	/26	10.0.0.65 – 10.0.0.126

Customers	10.0.0.128	255.255.255.192	/26	10.0.0.129 – 10.0.0.190
Admins	10.0.0.192	255.255.255.192	/26	10.0.0.193 – 10.0.0.254

5. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Constraint: Show your calculation steps and final answers.



IP Address - 172.16.0.0/16

original Subnet mask : 255.255.0.0

Req = 20

$2^n \geq \text{Required Subnet}$

$2^4 = 16$

$2^5 = 32$

So, we need = 20 of Subnet

Need to take 5 bits from next octet

255.255.248.0
11111111 11111111 11111000 00000000

→ calculate host per Subnet

$2^7 - 2$
 $2^11 - 2$

= 2048 - 2

= 2046 host per Subnet